

$y = f(x)$	Scalar function of one variable x – Independent variable (argument/input of a function) y – Independent variable (value of function)
$y = f(x_1, x_2)$	Scalar function of two variables x_1, x_2 – Independent variables (arguments/inputs of a function) y – Independent variable (value of function)
$\frac{dy}{dx}, y', f'(x)$	Derivative of first order of function $y = f(x)$
$\frac{d^2y}{dx^2}, y'', f''(x)$	Derivative of second order of function $y = f(x)$
$\frac{\partial y}{\partial x_i}, \frac{\partial f(x_1, x_2)}{\partial x_i}$	Partial derivatives of first order for function $y = f(x_1, x_2)$ with respect to variable x_i $i = 1, 2$
$\frac{\partial^2 f(x_1, x_2)}{\partial x_i \partial x_j}, \frac{\partial^2 (x_1, x_2)}{\partial x_i^2}, i = 1, 2$	Partial derivatives of second order for function $y = f(x_1, x_2)$
$H(x_1, x_2) = \begin{bmatrix} \frac{\partial^2 f(x_1, x_2)}{\partial x_1^2} & \frac{\partial^2 f(x_1, x_2)}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f(x_1, x_2)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x_1, x_2)}{\partial x_2^2} \end{bmatrix}$	Hessian, symmetric matrix of partial derivatives of 2 nd order for function $y = f(x_1, x_2)$
$\left. \frac{df(x)}{dx} \right _{x=\bar{x}}$	Value of a derivative of one variable function at point $\bar{x}$
$\left. \frac{\partial f(x_1, x_2)}{\partial x_i} \right _{\mathbf{x}=\bar{\mathbf{x}}}$	Value of a derivative of one variable function at point $\bar{\mathbf{x}} = (x_1, x_2)$
$\langle \mathbf{p}, \mathbf{x} \rangle = \sum_{i=1}^2 p_i x_i$	Scalar product of two vectors $\mathbf{p}, \mathbf{x} \in \mathbb{R}_+^2$
$\det(\mathbf{A})$ or $ \mathbf{A} $	Determinant of matrix $\mathbf{A}$
$d_E(\mathbf{x}^1, \mathbf{x}^2) = \left(\sum_{i=1}^2 (x_i^1 - x_i^2)^2 \right)^{\frac{1}{2}}$	Euclidean metric (distance)
$d(\mathbf{x}^1, \mathbf{x}^2) = \max_{i=1,2} \{ x_i^1 - x_i^2 \}$	Non-Euclidean metric (distance)
$\ \mathbf{x}_E\ = \left(\sum_{i=1}^2 (x_i)^2 \right)^{\frac{1}{2}}$	Euclidean norm
$\ \mathbf{x}\ = \max_{i=1,2} \{ x_i \}$	Non-Euclidean norm
$t = 0, 1, 2, \dots$	Time as discrete variable
$t \in [0; +\infty)$	Time as continuous variable
$\nabla f, \text{grad}f, \text{Gradient of function } f, \nabla f = (\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n})$	

**SOME LETTERS OF THE GREEK ALPHABET (AND
CORRESPONDING SYMBOLS) YOU MAY ENCOUNTER**

Symbol	Greek Alphabet	Symbol	Greek Alphabet
α	Alpha	θ	Theta
β	Beta	λ	Lamba
γ	Gamma	π	Pi
δ	Delta	σ	Sigma
ε	Epsilon	χ	Chi
ψ	Psi	μ	Mu
ρ	Rho	ω	Omega





BLOCK 1

REVIEW OF BASIC MATHEMATICS

Unit 1	Fundamental Concepts of Mathematics
Unit 2	Overview of Basic Methods of Mathematics
Unit 3	Relations and Functions
Unit 4	Co-ordinate Geometry and Representation of Functions



UNIT 1 FUNDAMENTAL CONCEPTS OF MATHEMATICS

Structure

- 1.0 Objectives
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 - 1.2.7.1 Difference between the Concepts of 'Relation' and 'Operation'
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- 1.3 Numbers: From Natural to Complex
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- 1.4 Decimal Representation of Rational and Irrational Numbers
- 1.5 Real Number Exponentiation
- 1.6 Let Us Sum Up
- 1.7 Key Words
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- 1.9 Exercises

1.0 OBJECTIVES

After going through this unit, you should be able to:

- differentiate between concept, term, notation and definition of number;
- explain the concept of set;
- use set notation;
- find relations on sets;
- apply set operations;

- differentiate between the concepts of ‘relation’ and ‘operation’ on sets;
- discuss relations and operations on numbers and their properties;
- define various types of numbers;
- apply operations on them; and use their properties in numerical problem solving;
- express real numbers, including irrational numbers, as decimal numbers;
- apply rules of exponentiation of numbers.

1.1 INTRODUCTION

The discussion in the unit is centred on the two foundational concepts of Mathematics, viz. (i) *number*, the basis of conventional mathematics; and (ii) *set*, the basis of modern mathematics.

Let us first understand the concept, term/name, notation/symbol, and system of the number.



Fig. 1 (a)



Fig. 1 (b)



Fig. 1 (c)

Number Concept: when one views Fig. 1(a) as *different* from Fig. 1(b), and views Fig. 1(b) as *similar* to Fig. 1(c), then one has number-sense, the idea/concept of number (of at least up to 2). The number sense/concept tells one that despite so many common features that both figures Fig. 1(a) & Fig. 1(b) have, e.g., both are about horse(s), and even about black horses, they are in one sense different. Similarly, about figures Fig. 1(b) & Fig. 1(c), despite having so many differences—Fig. 1(b) being picture of actual black horses, but Fig. 1(c) being picture of marble white elephants (toys)—the two figures have something in common, viz. ‘twoness’. The ‘twoness’ is a concept.

Number name/term: for the number of horses in Fig. 1(b), there are different names/terms in different languages: ‘two’ in English, ‘दो’ in Hindi, ‘zwei’ in German, ‘இரண்டு’ or ‘*Irantu*’ in Tamil.

Concept & Term: the word ‘term’ is used for the name of a concept. **For example, the terms** ‘two’ and ‘दो’ represent the same concept of ‘twoness’. For this concept, even in English, other terms like ‘pair’, ‘couple’, and even ‘twin’ may be used in some contexts. On the other hand, the **term** ‘bank’ may refer to two different **concepts**, viz. the one in context of a *river*, and the other in context of ‘*the place where money is deposited*’ etc.

Numeral/Number-Notation/Number-Symbol: For the number ‘six’, various known notations include ‘6’, ‘VI’, and ‘६’. Thus 6 is actually *name of a number*, i.e., it is numeral, and *not* a number. *But it is incorrectly called a number.*

In Mathematics, it is a practice to call the numerals as numbers. We will also generally use the term *number*, instead of the actual one the *numeral*.

Term, notation and definition: One of the mathematical *terms* is ‘square root’; the corresponding *notation/symbol* is $\sqrt{}$. And the *definition* of square-root is ‘For a given number x , its square-root y is a number which satisfies the equation $y \times y = x$, where the meaning of ‘ $\times$ ’ or multiplication is already known’

Numeral system (or system of numeration): We have heard of various *number* systems (more precisely, *numeral* systems), e. g., decimal number system, Roman numeral system, binary number system. For example, the decimal number ‘41’ may be represented as ‘*XLI*’ in Roman numeral system, and as ‘*101001*’ in binary number system. The same *expression* may represent different numbers in different number system. For example, the *expression* ‘*111*’ represents ‘*One hundred & eleven*’ in decimal number system, ‘*seven*’ in binary number system, and ‘*three*’ in ‘unary number system’.

The term ‘*Number System*’—against the term ‘*Numeral system*’—is different, and will be discussed later.

1.2 SET THEORY

Conventionally, the concept of *number* and geometric concepts like ‘*straight line*’, ‘*triangle*’ and ‘*square*’. have been the most fundamental, and the very first of the concepts discussed in introductory mathematics. Since almost the middle of the previous century, the concept of ‘*set*’ is considered as the most fundamental concept of mathematics and all other concepts, including even those of counting numbers and geometrical shapes, are discussed in terms of *set*.

1.2.1 Concept of Set

Set is a basic concept like ‘point’, & ‘straight line’ in geometry and like ‘number’ in arithmetic, and hence, should be left undefined. However, in view of the fact, that ‘set’ is now-a-days considered as one of the most fundamental concepts in mathematics; to make a beginning in *new* mathematics, we just give a sort of *working definition*, and define some elementary operations on sets to be frequently used in the subsequent discussion.

Roughly, a set may be considered as a *collection* of things or entities, like names of animals, names of books, names of numbers, names of geometrical shapes, or any mixture of these. Any entity in a set is called its *member*, or its *element*. Further, in a set, no member is repeated more than once. Also, the entities must be *distinctly identifiable*. For example, as the drops of water in a glass cannot be distinctly identified, therefore drops of water in a glass **do not** form a set.

More formally, we take a ***set as a well-defined collection of objects***. What is meant by ‘well-defined’ needs to be understood clearly. ***A collection is said to be well-defined, if (i) entities included in it are distinctly identifiable, and, (ii) if we***

pick up any distinctly identifiable entity in the world, then we should be able to tell whether the picked-up entity is in (i. e. belongs to) the collection or not. For example, let us consider the collection of **all** those students of IGNOU who enrolled in 2021 for M.A. (Economics) program. This is a well-defined collection, and hence, a set. But if we say the collection of **some of** those students of IGNOU who enrolled in 2021 for M. A. (Economics) program, **then, it is NOT a set.** Also, if we say ‘collection of all drops of water in this glass’, **then it is NOT a set,** because, as mentioned earlier, a ‘drop of water’ is not distinctly identifiable. Thus, **not any collection is a set, only a well-defined collection is a set.**

Let us look at a well-known example of a collection which is not well-defined, and hence is not a set:

A small village has only one (male) barber, who shaves all those, and only those males, who do not shave themselves. Let X be the **collection** of all the villagers shaven by the barber. Then, it is not possible to tell whether barber belongs to X or not. This is so because, if the *barber as a villager* shaves himself, then he cannot be shaven by the barber. Similarly, if *he, as a villager* does not shave himself, then he must be shaven by the barber. Thus, it is not possible to tell whether the barber is in X or not. Hence, X **is not a well-defined collection,** and thus, **not** a set.

1.2.2 Notations

Like most of the mathematical concepts, sets have special notation. For illustration, the **notation** $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is used to denote the set of all decimal digits, viz., 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

The notation for a set has at the leftmost position a *left brace*, denoted by ‘{’, followed by one of the elements, say 0, then a comma and a blank-space, next another element say 1, followed by a comma and blank-space, and so on, till the last element is written, which is followed by the *right-brace*, denoted by ‘}’. Generally, a set is given a name, e. g., in this case the name of the set may be **Num-Digits**, then we write $\text{Num-Digits} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Any *distinct* name—the name which has not been used earlier for naming any other set—may be assigned. However, generally, name assigned is *mnemonic* (i. e., name assigned should be helpful in remembering its contents or members).

Member/element of a set is what is included, except commas, within braces. For example, 0 is a member, or element, of Num-Digits, 5 is a member of Num-Digits etc. Instead, we may also say 0 **belongs to** Num-Digits, 5 **belongs to** Num-Digits etc. However, we say 35 **does not belong to** Num-Digits.

Notation for member is the symbol $\in$, called *epsilon*. It is used to denote membership of a set. For the above example of Num-Digits, we will write $0 \in \text{Num-Digits}$, $9 \in \text{Num-Digits}$ etc. Similarly, if we have a set called Vowels, with $\text{Vowels} = \{a, u, e, o, i\}$, then $e \in \text{Vowels}$, $u \in \text{Vowels}$ ETC. *Whenever x is a*

member of the set X , or equivalently, when $x \in X$, then we also say ' x belongs to X '. The notation for ' x is not a member of' or for ' x does not belong to' is $\notin$. For example, for the definition of *Num-Digits* given above, $35 \notin \text{Num-digits}$.

Convention for notation of sets and elements: generally, set names start with a capital letter, e. g. X , Y , N , *Num-Digits*, *Value*. And *members*, or *elements* are denoted by small letters, e. g. a , b , x , y etc. However, it is *not* at all binding; we may use small letters for set-names and capital letters for elements/members.

Extension of Set Notation: the problem with the set notation mentioned above, arises when the number of elements in a given set is either very large or infinite, because then it is quite difficult or even impossible to specify all the elements for using the notation. In such cases, if there is some pattern among the elements, then only some initial members are mentioned followed by **triple dots (...)**. For example, the set of numbers, each of which is less than 10,001 and is also square of some integer, may be denoted by $\{0, 1, 4, 9, \dots, 100, 121, \dots, 400, \dots, 10000\}$, or even more briefly by $\{0, 1, 4, 9, \dots, 10000\}$ etc. It may be named as *Sq_int_less_10001*.

1.2.3 Forms of Set Representation

The following two are well-known forms for denoting sets:

- i) **Tabular or roster form:** The notation, so far used for denoting a set, is called **tabular or roster form**. In this form, each of the elements/ members, or enough members to indicate a pattern must be included within braces. The notation is quite convenient if the number of members is a small number. However, the notation has problems, specially, in the case of large sets.
- ii) **Set-builder form:** The other notation is called **set-builder form**. Consider again the set *Sq_int_less_10001*. Every element of it has two properties: (a) it is square of an integer and (b) it is less than 10001.

The *set-builder form* uses these properties to denote the set as follows

$$Sq_int_less_10001 = \{x: x = y^2, y \text{ is an integer, and } x < 10001\},$$

The notation above states that the members of the set *Sq_int_less_10001* are x , where ('where' is denoted by ':' or '|') x satisfies three conditions: $x = y^2$, y is an integer, and $x < 10001$. There may be shades of differences in the set-builder notation. For example

The set *Decimal_digits* = $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ may be denoted as

$$Decimal_digits = \{x: x \in N \text{ and } x < 10\} \quad \text{OR}$$

$$Decimal_digits = \{x \in N: x < 10\} \text{ etc.}$$

The set-builder notation is quite convenient if the properties satisfied by all the members can be easily described, even if the set has very large number or infinite number of members.